

PRACHI SINGH

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EDUCATION

University of Illinois Urbana-Champaign Master of Computer Science	Aug 2024 – Dec 2025
Relevant Coursework – Algorithms, Distributed Algorithms, Database Systems, Software Engineering, Cloud Storage Systems, User-Centered Machine Learning	GPA – 3.44/4
SRM Institute of Science and Technology Bachelor of Technology in Computer Science Engineering	July 2018 – May 2022
Relevant Coursework - Data Structures, Distributed Operating Systems, Database Security and Privacy	CGPA – 9.53/10

WORK EXPERIENCE

Neuro Leap Corporation Software Engineer Intern	Feb 2026 – Present
- Developed full-stack features for the VividKids blog platform using React and Node.js, including a post/comment reaction system, user authentication middleware, role-based access control, and profile management with persistent image storage. - Building PostgreSQL schemas and REST API endpoints for core platform entities including users, posts, comments, reactions, and profiles, integrating JWT-based authorization and server-side pagination to support scalable content delivery.	

COUNTRY Financial® Software Development Intern	May 2025 – Dec 2025
- Engineered an LLM-driven automation framework using Azure AI Foundry to convert legacy HQL queries to PySpark, achieving 90%+ code accuracy via advanced prompt engineering and reducing manual refactoring effort by 80%. - Led development of an internal tool using React and Django to automate test data generation, improving developer productivity by reducing manual processing time from 1 week to under 5 minutes. - Architected claims automation tool for Guidewire platform integrating LangChain, Azure OpenAI, and Azure Document Intelligence, eliminating manual email triage, saving 2,000+ annual labor hours and reducing claim processing time.	

JP Morgan Chase & Co. Software Engineer I	Aug 2022 – July 2024
- Built and deployed a Redis cache-aside layer with appropriate TTLs and eviction policies, cutting database queries by ~50% and reducing API latency for high-traffic endpoints. - Migrated data ingestion pipelines from on-premises file systems to AWS S3, updating application configurations and data layer logic to support cloud-native architecture, enabling ~40% faster data retrieval. - Orchestrated multiple complex ETL workflows using Python and Cron in Athena to automate data generation and publication, eliminating manual triggers and cutting delivery delays by 90%. - Automated cross-platform data synchronization using Java Spring Boot REST APIs, eliminating 80% of manual syncing effort across multiple platforms. - Resolved P1 production incidents within 15-minute SLA, utilizing Splunk for log analysis and Grafana for system health monitoring to perform RCA, achieving ~99.9% uptime for internal front-office applications serving 500+ users.	

PROJECTS

Performance Benchmarking of Distributed Storage Systems for IoT	Oct 2025 – Dec 2025
- Benchmarked distributed shared logging architectures (Kafka, Pulsar, RabbitMQ) for real-time IoT telemetry stream, analyzing end to end throughput vs. latency trade-offs to determine the optimal data ingestion architecture. - Simulated high-velocity sensor streams using Python, implementing batch processing strategies to stress-test system limits and evaluate write durability and consistency guarantees under peak load.	

Carpe Diem	Oct 2025 – Dec 2025
- Developed a full MERN-stack productivity platform with time-blocking and gamified rewards, hosted on Vercel and Render, supporting multiple goal modes (time management, productivity, procrastination). - Implemented scalable backend services using Node.js, Express and MongoDB, integrating Passport.js, OAuth2 and Cron jobs to automate task state transitions, reward calculation, analytics and notification workflows.	

TECHNICAL SKILLS

- Programming Languages:** Java, Python
- AI / LLMs :** LangChain, Azure AI Foundry
- Frontend :** HTML, CSS, React.js, Vue.js, JavaScript, TypeScript, Tailwind
- Backend :** Node.js, Django, Spring Boot, GraphQL, RESTful APIs, OAuth, OIDC, JWT
- Database Management :** MySQL, PostgreSQL, Oracle, PySpark, MongoDB, Redis
- Cloud & DevOps :** AWS (EC2, S3, Lambda), Docker, CI/CD (Jenkins, GitLab)
- Tools :** Git, Linux, Jira, Github Copilot, IntelliJ IDEA, Android Studio, Swagger, Splunk

LEADERSHIP & ACHIEVEMENTS

- Secured 1st position at UIUC Research Park Hackathon 2025 for PartyMode, a real-time collaborative music app that solves the “aux cord problem” at parties by democratizing music selection. Built multi-user voting, live playlist synchronization, and admin controls using React.js, Tailwind and Spotify API, enabling seamless crowd-sourced playlists.
- Mentored summer interns at JPMC, providing technical guidance and conducting code reviews to accelerate their onboarding and ensure high-quality project delivery.